THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-B-3990

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

Form Approved
OMB No. 0704-0188

INCH DESIGN

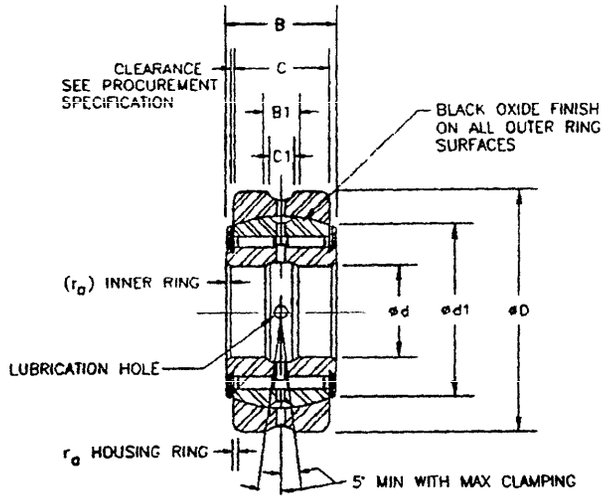



TABLE I — DIMENSIONS IN INCHES

| Dash No. | ød Bore | øD Housing Ring Outside Dia | D Overall Width | C Outer Ring Width | ød1 Washer Outside Dia | B1 Lubrication Groove Width | C1 Lubrication Groove Width | $r_a$ 1/ Fillet Max | Total Radial Clearance Max. | $d_a$ Clamping Diameter Max | $d_a$ Clamping Diameter Min | 2/ Limit Load Rating lbf | Mass (Approx) lbf | Housing Size Gage +0.0000 −0.0001 Low | Housing Size Gage +0.0000 −0.0001 High |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| −6 | 0.3750 | 1.1875 | 0.562 | 0.469 | 0.812 | 0.188 | 0.125 | 0.022 | 0.0030 | 0.781 | 0.641 | 4530 | 0.130 | 1.1867 | 1.1872 |
| −7 | 0.4375 | 1.3125 | 0.625 | 0.531 | 0.875 | 0.188 | 0.125 | 0.032 | 0.0031 | 0.844 | 0.703 | 5870 | 0.174 | 1.3116 | 1.3122 |
| −8 | 0.5000 | 1.5000 | 0.750 | 0.656 | 1.031 | 0.188 | 0.125 | 0.032 | 0.0031 | 1.000 | 0.844 | 8670 | 0.293 | 1.4991 | 1.4997 |
| −9 | 0.5625 | 1.6875 | 0.875 | 0.781 | 1.093 | 0.188 | 0.156 | 0.032 | 0.0031 | 1.062 | 0.891 | 11800 | 0.42 | 1.6866 | 1.6872 |
| −10 | 0.6250 | 1.7500 | 1.000 | 0.906 | 1.156 | 0.250 | 0.156 | 0.032 | 0.0032 | 1.094 | 0.953 | 15500 | 0.52 | 1.7491 | 1.7497 |
| −12 | 0.7500 | 1.8750 | 1.125 | 1.000 | 1.281 | 0.250 | 0.156 | 0.032 | 0.0034 | 1.156 | 1.070 | 20000 | 0.63 | 1.8741 | 1.8747 |
| −14 | 0.8750 | 2.1250 | 1.250 | 1.125 | 1.500 | 0.375 | 0.156 | 0.032 | 0.0037 | 1.375 | 1.250 | 25800 | 0.87 | 2.1238 | 2.1246 |
| −16 | 1.0000 | 2.2500 | 1.250 | 1.125 | 1.625 | 0.375 | 0.156 | 0.032 | 0.0041 | 1.500 | 1.375 | 28700 | 0.96 | 2.2488 | 2.2496 |
| −20 | 1.2500 | 2.5000 | 1.250 | 1.049 | 1.906 | 0.375 | 0.156 | 0.032 | 0.0041 | 1.781 | 1.625 | 31400 | 1.07 | 2.4988 | 2.4996 |
| −24 | 1.5000 | 2.7500 | 1.250 | 1.049 | 2.156 | 0.375 | 0.156 | 0.032 | 0.0041 | 2.062 | 1.875 | 36600 | 1.23 | 2.7488 | 2.7496 |
| −32 | 2.0000 | 3.2500 | 1.250 | 1.049 | 2.656 | 0.375 | 0.156 | 0.032 | 0.0045 | 2.593 | 2.375 | 47100 | 1.49 | 3.2496 | 3.2496 |
| −40 | 2.5000 | 3.7500 | 1.250 | 1.049 | 3.154 | 0.375 | 0.156 | 0.032 | 0.0046 | 3.062 | 2.875 | 57500 | 1.78 | 3.7485 | 3.7495 |
| −48 | 3.0000 | 4.2500 | 1.250 | 1.049 | 3.656 | 0.375 | 0.156 | 0.032 | 0.0055 | 3.562 | 3.375 | 67900 | 2.06 | 4.2485 | 4.2495 |
| −56 | 3.5000 | 4.8750 | 1.250 | 1.049 | 4.219 | 0.375 | 0.156 | 0.044 | 0.0057 | 4.141 | 3.969 | 80100 | 2.65 | 4.8735 | 4.8745 |

TABLE II — DIMENSIONS IN MILLIMETERS

| Dash No. | ød Bore | øD Housing Ring Outside Dia | D Overall Width | C Outer Ring Width | ød1 Washer Outside Dia | B1 Lubrication Groove Width | C1 Lubrication Groove Width | $r_a$ 1/ Fillet Max | Total Radial Clearance Max. | $d_a$ Clamping Diameter Max | $d_a$ Clamping Diameter Min | 2/ Limit Load Rating N | Mass (Approx) Kg | Housing Size Gage +0.0000 −0.0003 Low | Housing Size Gage +0.0000 −0.0003 High |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| −6 | 9.525 | 30.162 | 14.27 | 11.91 | 20.62 | 4.78 | 3.18 | 0.6 | 0.076 | 19.84 | 16.28 | 20100 | 0.059 | 30.142 | 30.155 |
| −7 | 11.112 | 33.338 | 15.88 | 13.49 | 22.22 | 4.78 | 3.18 | 0.8 | 0.079 | 21.44 | 17.86 | 26100 | 0.079 | 33.315 | 33.330 |
| −8 | 12.700 | 38.100 | 19.05 | 16.66 | 26.19 | 4.78 | 3.18 | 0.8 | 0.079 | 25.40 | 21.44 | 38500 | 0.133 | 38.079 | 38.092 |
| −9 | 14.288 | 42.862 | 22.22 | 19.04 | 27.76 | 4.78 | 3.96 | 0.8 | 0.079 | 26.97 | 22.63 | 52300 | 0.19 | 42.840 | 42.855 |
| −10 | 15.875 | 44.450 | 25.40 | 23.01 | 29.36 | 6.35 | 3.96 | 0.8 | 0.081 | 27.79 | 24.21 | 68700 | 0.24 | 44.427 | 44.442 |
| −12 | 19.050 | 47.625 | 28.58 | 25.40 | 32.54 | 6.35 | 3.96 | 0.8 | 0.086 | 29.36 | 27.38 | 88700 | 0.29 | 47.602 | 47.617 |
| −44 | 22.225 | 53.975 | 31.75 | 28.58 | 38.10 | 9.52 | 3.96 | 0.8 | 0.094 | 34.92 | 31.75 | 115000 | 0.39 | 53.945 | 53.965 |
| −46 | 25.400 | 57.150 | 31.75 | 28.58 | 41.28 | 9.52 | 3.96 | 0.8 | 0.104 | 38.10 | 34.92 | 127000 | 0.44 | 57.120 | 57.140 |
| −20 | 31.750 | 63.500 | 31.75 | 26.64 | 48.41 | 9.52 | 3.96 | 0.8 | 0.104 | 45.24 | 41.28 | 140000 | 0.49 | 63.470 | 63.490 |
| −24 | 38.100 | 69.850 | 31.75 | 26.64 | 54.76 | 9.52 | 3.96 | 0.8 | 0.104 | 52.37 | 47.62 | 163000 | 0.56 | 69.820 | 69.840 |
| −32 | 50.800 | 82.550 | 31.75 | 26.64 | 67.46 | 9.52 | 3.96 | 0.8 | 0.114 | 65.86 | 60.32 | 209000 | 0.68 | 82.512 | 82.540 |
| −40 | 63.500 | 95.250 | 31.75 | 26.64 | 80.16 | 9.52 | 3.96 | 0.8 | 0.117 | 77.77 | 73.02 | 255000 | 0.81 | 95.212 | 95.237 |
| −48 | 76.200 | 107.950 | 31.75 | 26.64 | 92.86 | 9.52 | 3.96 | 0.8 | 0.142 | 90.47 | 85.72 | 302000 | 0.94 | 107.912 | 107.937 |
| −56 | 88.900 | 123.825 | 31.75 | 26.64 | 107.16 | 9.52 | 3.96 | 1.1 | 0.145 | 105.18 | 100.81 | 357000 | 1.20 | 123.787 | 123.812 |

1/ The chamfer on bearings must clear the maximum fillet radius given in the table. This specification does not control bearing chamfer contours

2/ The limit load rating can be defined as the maximum radial load which can be applied to a bearing without impairing the subsequent functioning of the bearing in airframe applications.
The ultimate or static fracture load rating is not less than 1.5 times the limit load rating.

(H) Denotes Change(s)

INCH-POUND

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY- AT NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AF-84 DLA-IS
USER:
PROJECT NUMBER: 3110-0858

MILITARY SPECIFICATION SHEET

TITLE
BEARING, ROLLER, NEEDLE-DOUBLE ROW, HEAVY DUTY, SELF-ALIGNING, TYPE IV, ANTIFRICTION, INCH

SPECIFICATION SHEET NUMBER
MS24464 05 DEC 94 REV H

SUPERSEDING
MS24464G 17 JUN 77

AMSC- N/A FSC 3110

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Form Approved
OMB No 0704-0188

TABLE III TOLERANCE LIMITS

DIMENSIONS IN INCHES

| ød Basic Bore | | Allowable Deviation from d of Single Mean DIA, d | | Allowable Deviation From Over-all Width, B | | Allowable Deviation From Washer Outside Dia., $d_1$ | | Allowable Deviation From Lubrication Groove Width, $B_1$ | | ØD Basic Outer Ring Outside Dia. | | Allowable Deviation From D of Single Mean Dia., D | | Allowable Deviation From Outer Ring Width, C | | Allowable Deviation From Lubrication Groove Width, $C1_1$ | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| OVER | INCL | HIGH | LOW | HIGH | LOW | HIGH | LOW | HIGH | LOW | OVER | INCL | HIGH | LOW | HIGH | LOW | HIGH | LOW |
| 0.3125 | 3.0000 | +0 | -0.0007 | +0 | -0.005 | +0.010 | -0.010 | +0.031 | -0.031 | 1.0625 | 1.8750 | +0 | -0.0005 | +0 | -0.005 | +0 | -0.062 |
| 3.0000 | 3.5000 | +0 | -0.0006 | +0 | -0.005 | +0.010 | -0.010 | +0.031 | -0.031 | 1.8750 | 2.7500 | +0 | -0.0006 | +0 | -0.005 | +0 | -0.062 |
| | | | | | | | | | | 2.7500 | 4.2500 | +0 | -0.0008 | +0 | -0.005 | +0 | -0.062 |
| | | | | | | | | | | 4.2500 | 4.8750 | +0 | -0.0010 | +0 | -0.005 | +0 | -0.062 |

TABLE IV. TOLERANCING LIMITS

DIMENSIONS IN MILLIMETERS

| ød Basic Bore | | Allowable Deviation From d of Single Mean Dia, d | | Allowable Deviation From Over-all Width E | | Allowable Deviation From Washer Outside Dia., $d_1$ | | Allowable Deviation From Lubrication Groove Width, B1 | | ØD Basic Outer Ring Outside Dia. | | Allowable Deviation From D of Single Mean Dia., D | | Allowable Deviation From Outer Ring Width C | | Allowable Deviation From Lubrication Groove Width, $C_1$ | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| OVER | INCL | HIGH | LOW | HIGH | LOW | HIGH | LOW | HIGH | LOW | OVER | INCL | HIGH | LOW | HIGH | LOW | HIGH | LOW |
| 7.938 | 76.200 | +0 | -.018 | +0 | -.13 | 40.25 | -.25 | +.5 | -.8 | 26.968 | 47.625 | +0 | -.013 | +0 | -.13 | +0 | -1.6 |
| 76.200 | 88.900 | +0 | -.020 | +0 | -.13 | 40.25 | -.25 | +.5 | -.8 | 47.623 | 69.850 | +0 | -.015 | +0 | -.13 | +0 | -1.6 |
| | | | | | | | | | | 69.850 | 107.950 | +0 | -.020 | +0 | -.13 | +0 | -1.6 |
| | | | | | | | | | | 107.950 | 123.825 | +0 | -.025 | +0 | -.13 | +0 | -1.6 |

TABLE V. OIL HOLE DATA

| BORE-DASH NO. | | NUMBER OF HOLES | | |
|---|---|---|---|---|
| OVER | INCL | INNER RING | OUTER RING | SPHERICAL HOUSING |
| 5 | 10 | 2 | 4 | 2 |
| 10 | 56 | 4 | 4 | 2 |

REQUIREMENTS:

(H) 1. MATERIAL: Steel, MIL-S-8690, ASTM A304, ASTM A576, ASTM A675, QQ-S-700, FED STD. NO.66, AISI/SAE Steel No 50100, 51100, 52100

(H) 2. PLATING: Zinc-Nickel plate in accordance with AMS 2417, Type 2, or cadmium plated in accordance with QQ-P-416, type II, class 2, with a thickness of .0003 inches, to .0006 inches.

(H) 3. MACHINE FINISH: ANSI/ASME B46:1, See Procurement Specification.

4. LUBRICATION: Grease conforming to MIL-G-81322.

5. Dimensions to be met after plating.

6. Remove all burrs and sharp edges.

NOTES:

1. This standard takes precedence over any other document referenced herein.

2. Referenced documents are the issue in effect at the date of invitation for bid.

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY- AT NAVY- AS
AIR FORCE- 99 DLA-
REVIEW: AF-84 DLA-IS
USER:
PROJECT NUMBER: 3110-0858

MILITARY SPECIFICATION SHEET

TITLE

BEARING, ROLLER, NEEDLE-DOUBLE ROW, HEAVY DUTY, SELF-ALIGNING, TYPE IV, ANTIFRICTION, INCH

SPECIFICATION SHEET NUMBER
MS24464 05 DEC 94 REV H

SUPERSEDING
MS24464G 17 JUN 77

AMSC- N/A FSC 3110

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